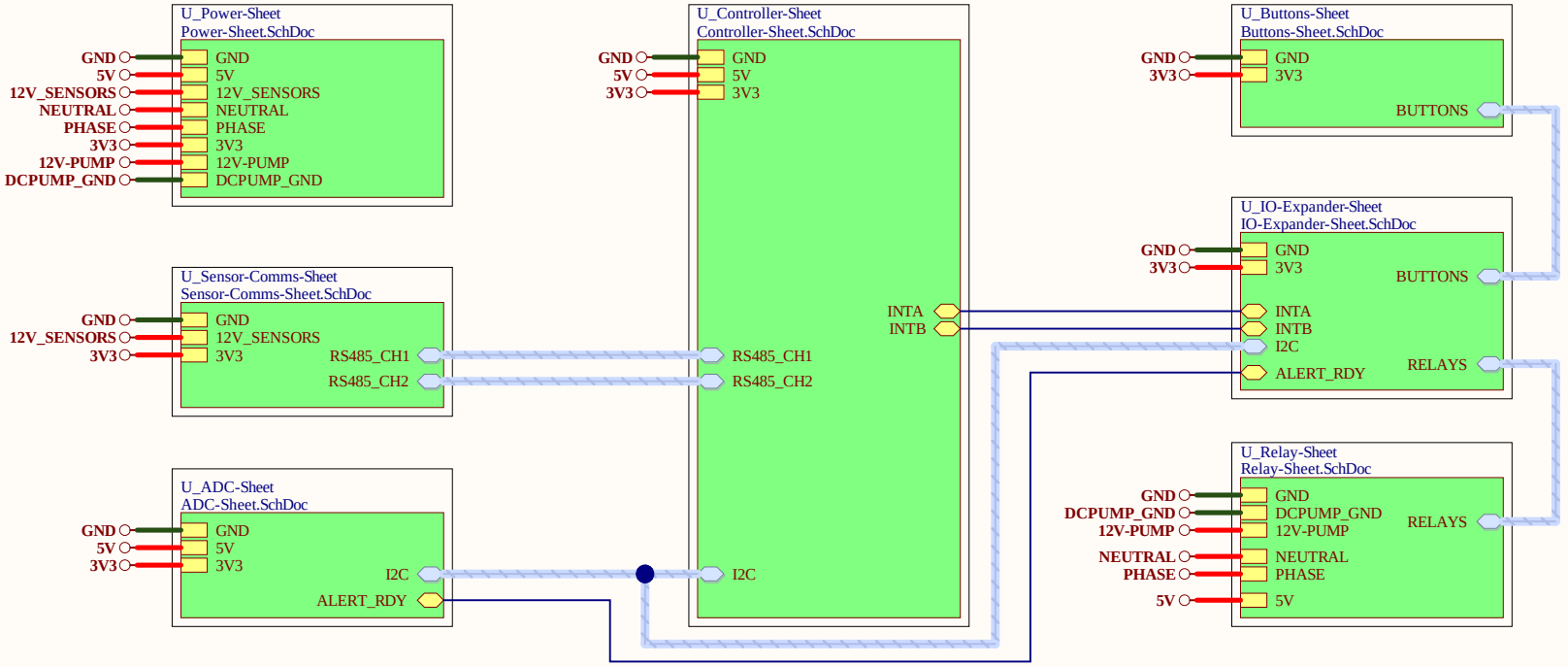
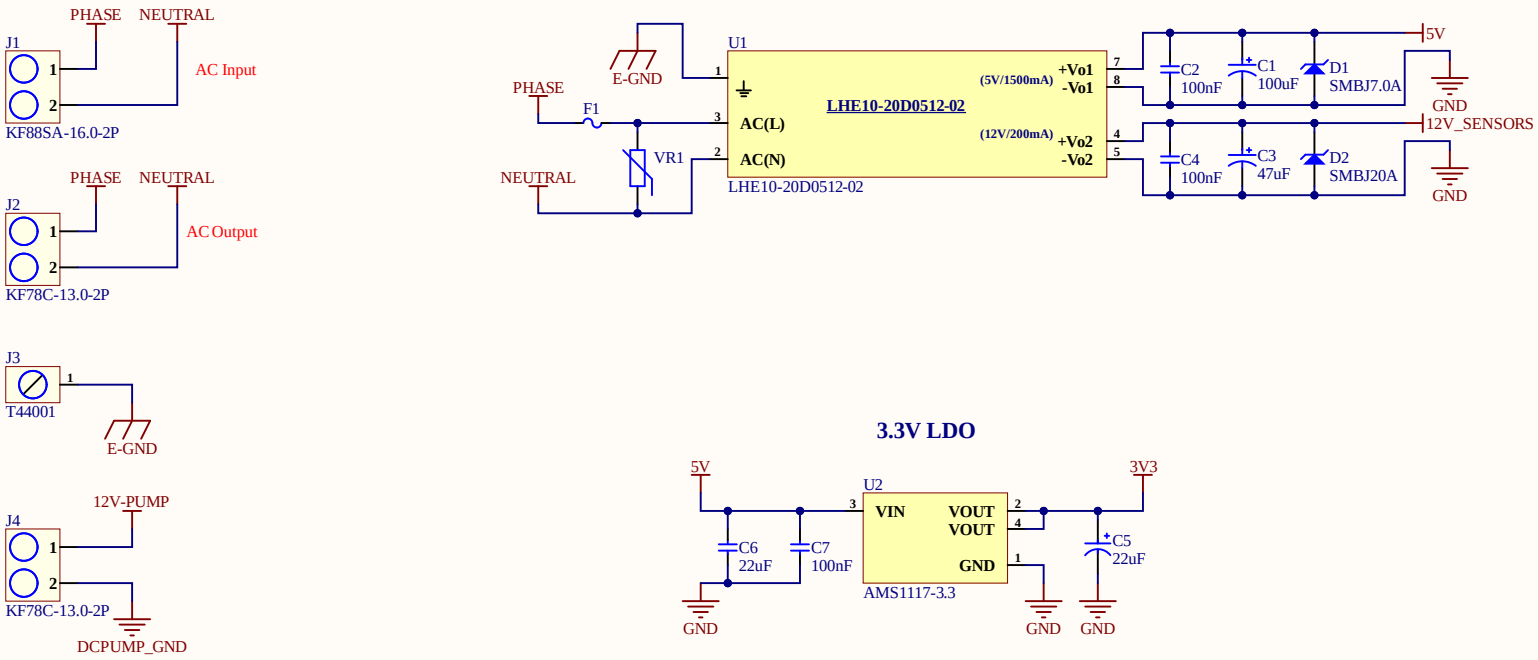


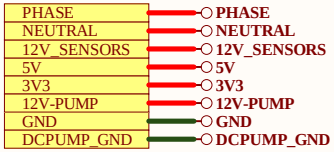
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File: LogNController-v1.1.PrjPcb	
Date: 8/7/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8

AC to DC Supply Module

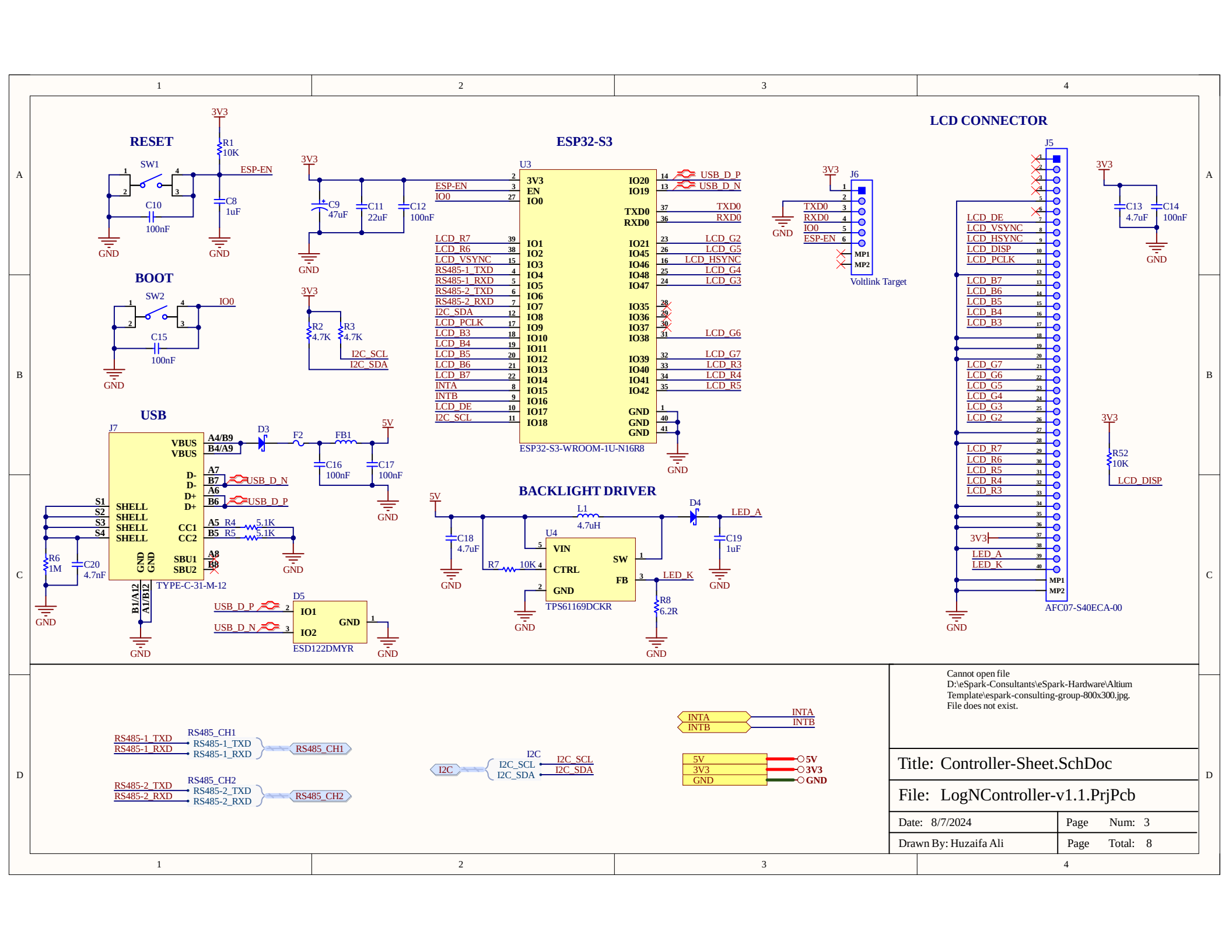


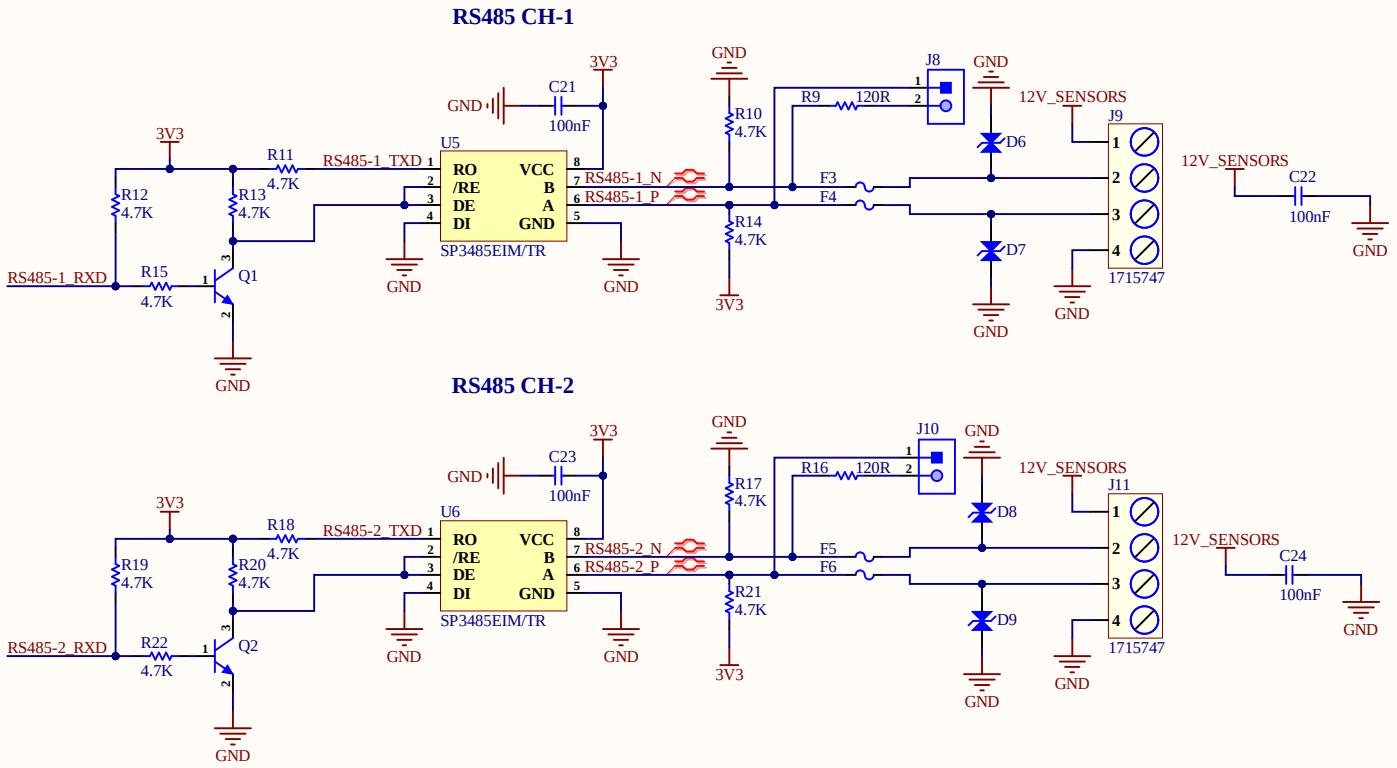
3.3V LDO

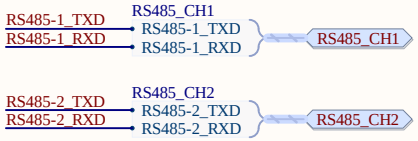


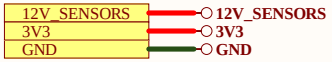
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Date: 8/7/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8


RESET
SW1
R1
10K
ESP-EN
C10
100nF
C8
1uF
BOOT
SW2
IO0
C15
100nF
USB
J7
TYPE-C-31-M-12
D3
F2
FB1
C16
100nF
C17
100nF
R6
1M
C20
4.7nF
R4
5.1K
R5
5.1K
D5
ESD122DMYR
ESP32-S3
U3
ESP32-S3-WROOM-1U-N16R8
C9
47uF
C11
22uF
C12
100nF
R2
4.7K
R3
4.7K
I2C_SCL
I2C_SDA
USB_D_P
USB_D_N
TXD0
RXD0
LCD_G2
LCD_G5
LCD_HSYNC
LCD_G4
LCD_G3
LCD_G6
LCD_G7
LCD_R3
LCD_R4
LCD_R5
BACKLIGHT DRIVER
L1
4.7uH
D4
LED_A
LED_K
C18
4.7uF
C19
1uF
R7
10K
R8
6.2R
U4
TPS61169DCKR
Voltlink Target
J6
LCD CONNECTOR
J5
AFC07-S40ECA-00
C13
4.7uF
C14
100nF
R52
10K
LCD_DISP
LCD_DE
LCD_VSYNC
LCD_HSYNC
LCD_DISP
LCD_PCLK
LCD_B7
LCD_B6
LCD_B5
LCD_B4
LCD_B3
LCD_R7
LCD_R6
LCD_R5
LCD_R4
LCD_R3
RS485-1_TXD
RS485-1_RXD
RS485_CH1
RS485-2_TXD
RS485-2_RXD
RS485_CH2
INTA
INTB
I2C
5V
3V3
GND
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File: LogNController-v1.1.PrjPcb
Date: 8/7/2024
Page Num: 3
Drawn By: Huzaifa Ali
Page Total: 8







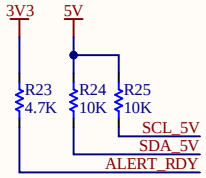
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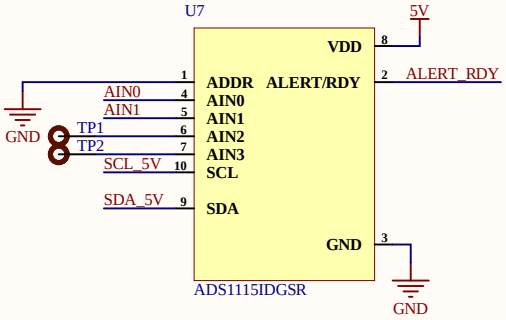
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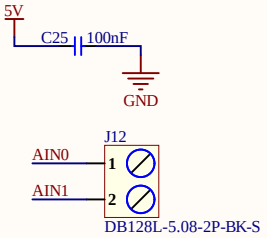
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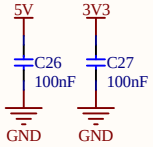
Date: 8/7/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8

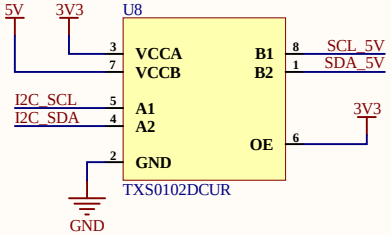
16-BIT ADC

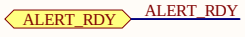




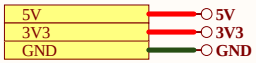








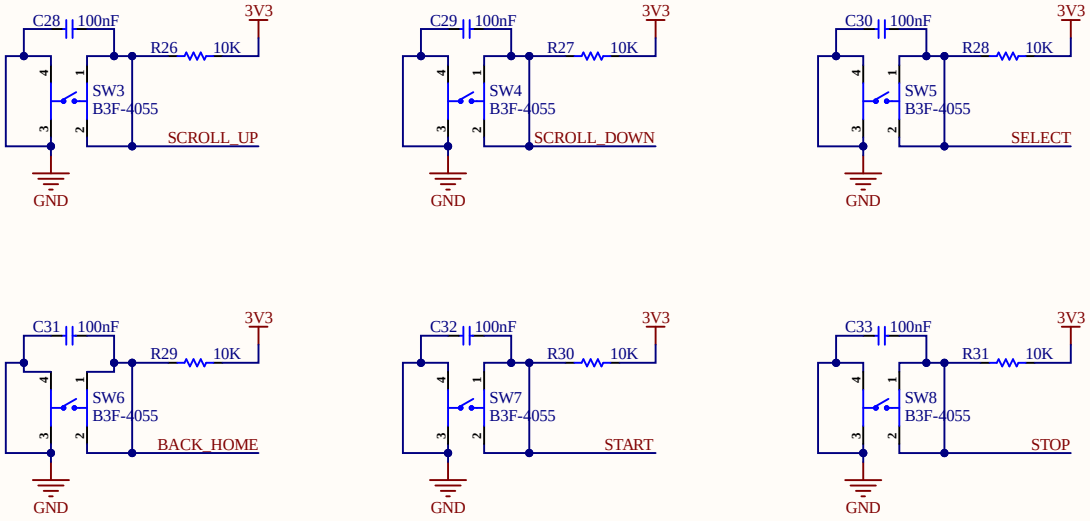


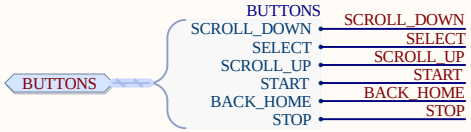


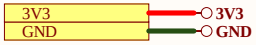
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Date: 8/7/2024	Page Num: 5
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BUTTONS



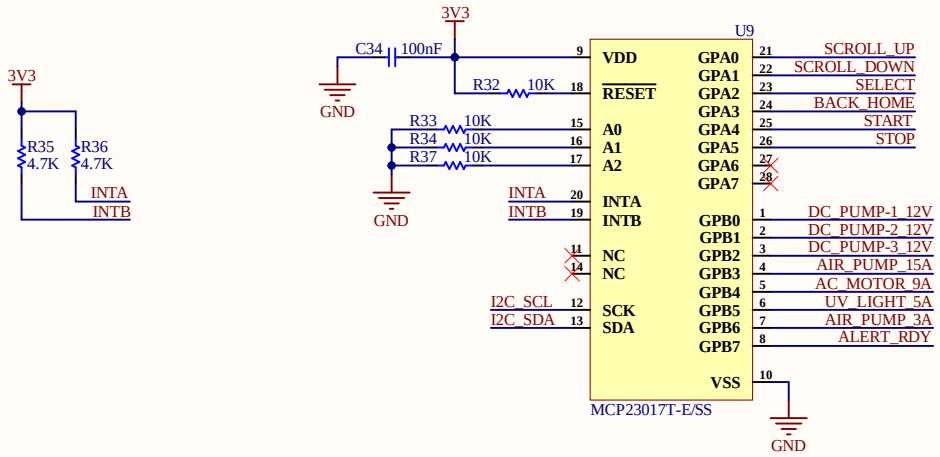


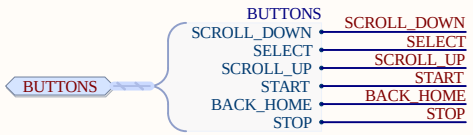


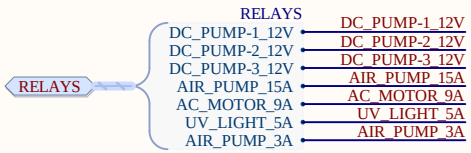
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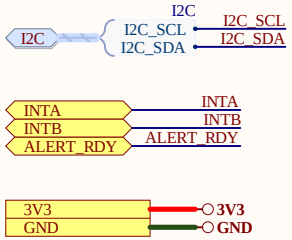
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File: LogNController-v1.1.PrjPcb	
Date: 8/7/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 8

IO Expander



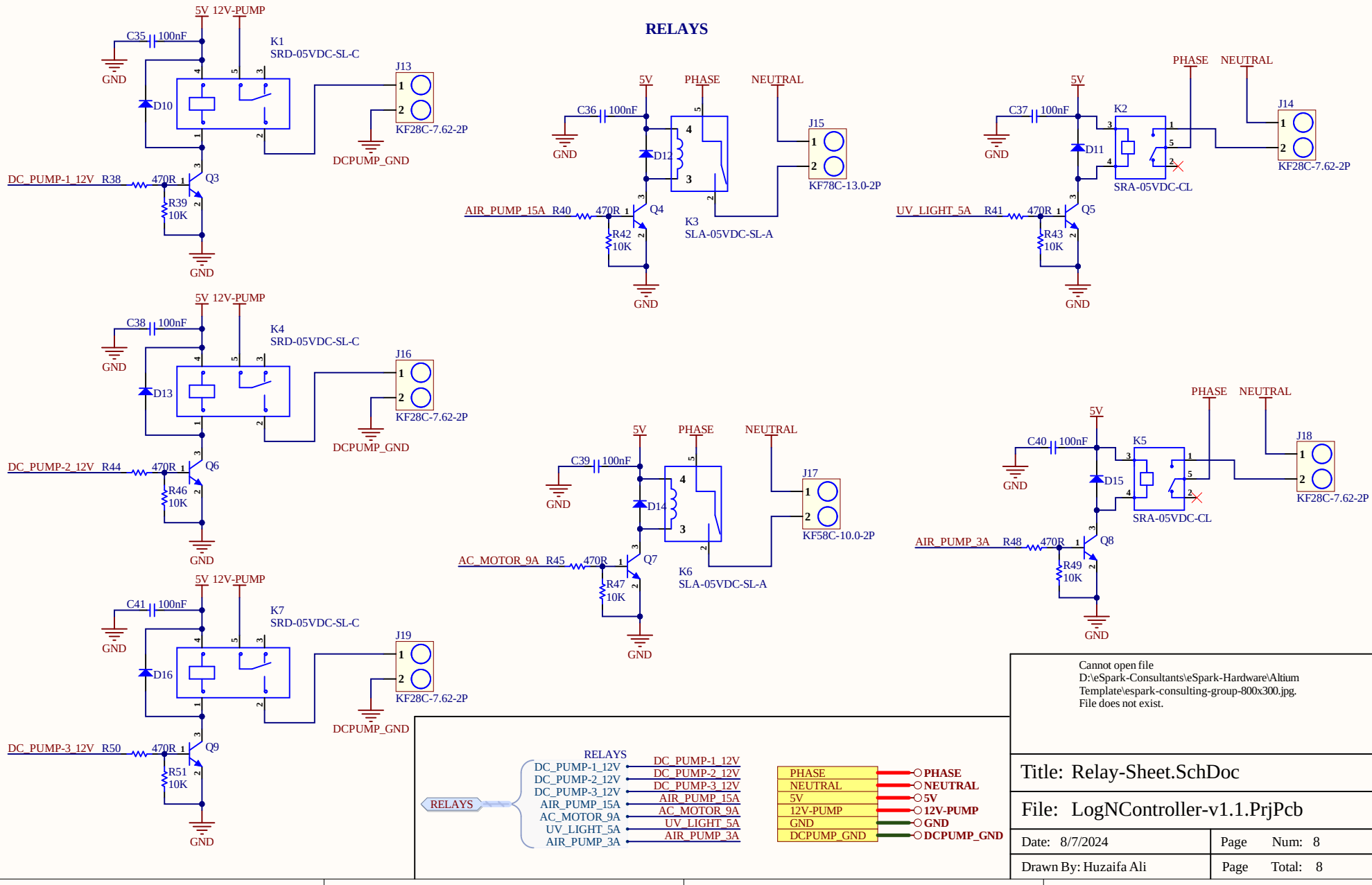


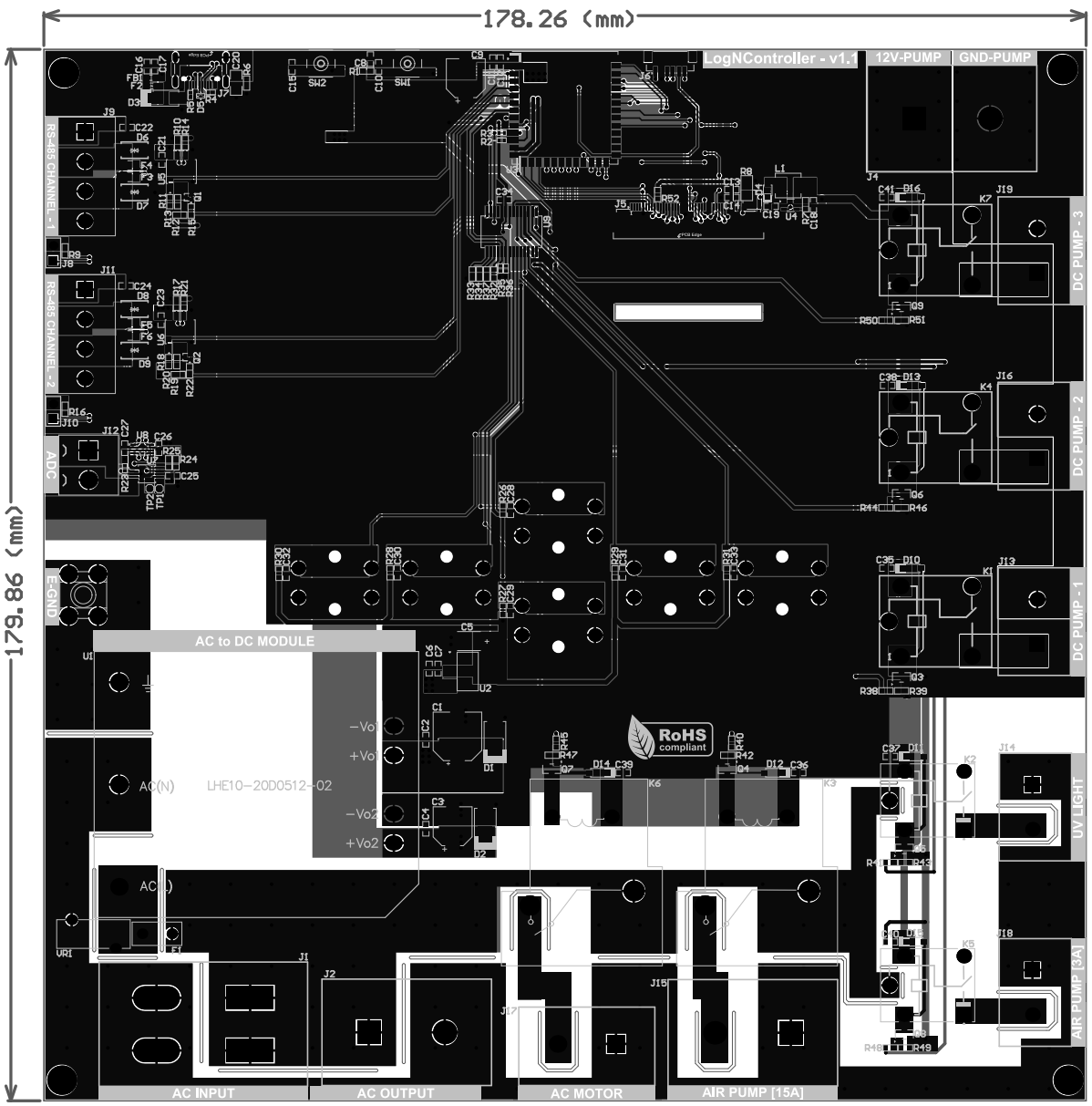




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Title: IO-Expander-Sheet.SchDoc	
File: LogNController-v1.1.PrjPcb	
Date: 8/7/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 8


RELAYS
5V 12V-PUMP
C35 100nF
K1
SRD-05VDC-SL-C
GND
D10
J13
KF28C-7.62-2P
DCPUMP_GND
DC_PUMP-1_12V R38 470R
Q3
R39 10K
GND
5V PHASE NEUTRAL
C36 100nF
D12
J15
KF78C-13.0-2P
AIR_PUMP_15A R40 470R
Q4
K3
SLA-05VDC-SL-A
R42 10K
GND
PHASE NEUTRAL
5V
C37 100nF
K2
J14
KF28C-7.62-2P
D11
SRA-05VDC-CL
UV_LIGHT_5A R41 470R
Q5
R43 10K
GND
5V 12V-PUMP
C38 100nF
K4
SRD-05VDC-SL-C
D13
J16
KF28C-7.62-2P
DCPUMP_GND
DC_PUMP-2_12V R44 470R
Q6
R46 10K
GND
5V PHASE NEUTRAL
C39 100nF
D14
J17
KF58C-10.0-2P
AC_MOTOR_9A R45 470R
Q7
K6
SLA-05VDC-SL-A
R47 10K
GND
PHASE NEUTRAL
5V
C40 100nF
K5
J18
KF28C-7.62-2P
D15
SRA-05VDC-CL
AIR_PUMP_3A R48 470R
Q8
R49 10K
GND
5V 12V-PUMP
C41 100nF
K7
SRD-05VDC-SL-C
D16
J19
KF28C-7.62-2P
DCPUMP_GND
DC_PUMP-3_12V R50 470R
Q9
R51 10K
GND
RELAYS
DC_PUMP-1_12V
DC_PUMP-2_12V
DC_PUMP-3_12V
AIR_PUMP_15A
AC_MOTOR_9A
UV_LIGHT_5A
AIR_PUMP_3A
PHASE
NEUTRAL
5V
12V-PUMP
GND
DCPUMP_GND
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File: LogNController-v1.1.PrjPcb
Date: 8/7/2024
Page Num: 8
Drawn By: Huzaifa Ali
Page Total: 8



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.39mil	3.5	
1	Top Layer		1.38mil		
	Dielectric 1	FR-4	59.06mil	4.8	
2	Bottom Layer		1.38mil		
	Bottom Solder	Solder Resist	0.39mil	3.5	
	Bottom Overlay				
Total board thickness:			62.60mil		